GBT Enterprise DashCam

TruckerCloud Integration API Reference Guide

Direct Telematics, ADAS/DMS Safety Events, Historical Breadcrumbs, and On-Demand Video Retrieval

Target Audience: TruckerCloud Integration Team	**Base URL:** https://api.gbttelematics.com (Production: https://3.85.81.10)
Auth Header: X-API-KEY: [Secure Key] (Sandbox: gbt-tc-sandbox-key-7a3e5c12)	**OpenAPI Docs:** https://api.gbttelematics.com/swagger-ui/index.html
Compliance: 17-char VIN, ECU Odometer, UTC Timezone, Metric km/h	**Hardware:** WanWay H20P 4G LTE Dual-Cam (Road + Cabin)

1. System Overview & Architecture

GBT operates an independent DashCam and telematics platform connecting directly to physical WanWay H20P dual-camera hardware via JT/T 808 (telemetry/commands on port 7800) and JT/T 1078 (RTP media transfer on port 10780). All continuous video footage records locally onto on-device high-endurance MicroSD cards to eliminate unnecessary cloud egress and storage overhead. Selected incident clips (harsh events, collisions, DMS driver alerts) and dispatcher-requested recordings are indexed and made available on-demand through these certified REST APIs.

Standard	Implementation Specification
Vehicle Identification	Strictly enforced 17-character ISO 3779 VIN with valid letters/digits (excluding I, O, Q).
Odometer Data	Always labeled with provenance: ECU (direct vehicle bus) or GPS_ESTIMATED. Units in kilometers.
Time Synchronization	All timestamps are formatted in UTC ISO 8601 (e.g., 2026-11-04T10:30:00Z). Telemetry includes explicit timezone field.
Speed Units	Strictly kilometers per hour (km/h) across all endpoints, exposed as both speedKmH and speed.
Authentication & Sandbox	Header X-API-KEY. Separate Sandbox (gbt-tc-sandbox-key-7a3e5c12) and Production keys provided securely.
Data Retention	High-resolution GPS telemetry retained for a minimum of 180 days (exceeds 90-day requirement).
Video Streaming	HLS format (m3u8 index with MPEG-TS segments) for both Channel 1 (Road) and Channel 2 (Cabin).

2. Certified TruckerCloud Integration Endpoints

GET /api/v1/integrations/truckercloud/auth/validate — Authentication & Environment Context Validation

Function & Purpose: *Validates the integration X-API-KEY and returns the authenticated service account, role, and operational environment context.*
Allows TruckerCloud systems to programmatically verify credential validity and confirm whether connected to SANDBOX or PRODUCTION.

Parameter / Header	Type	Description
X-API-KEY	*Header*	Required. Integration secret key (Sandbox: gbt-tc-sandbox-key-7a3e5c12)

Real Response Payload (Sanitized Sample):


```
{ "status": "VALID", "accountId": "srv-truckercloud-sandbox", "email": "sandbox@truckercloud.com", "role": "SUPER_ADMIN", "tenantId": "tenant-inurum", "environment": "SANDBOX", "timestamp": "2026-11-13T06:38:45Z" }
```


GET /api/v1/integrations/truckercloud/vehicles — Fleet Vehicles Master Registry

Function & Purpose: *Returns all active fleet vehicles mapped with validated 17-character VIN, device IMEI, assigned driver ID, and ECU odometer.*
TruckerCloud uses this endpoint to synchronize vehicle assets, discover newly installed DashCams, and associate telematics.

Parameter / Header	Type	Description
X-API-KEY	*Header*	Required. Server-to-server key (Sandbox: gbt-tc-sandbox-key-7a3e5c12)
page	*Query*	Optional. Zero-based page number (default: 0)
size	*Query*	Optional. Number of items per page (default: 50, max: 200)

Real Response Payload (Sanitized Sample):


```
{ "items": [ { "id": "veh-aebf556b", "vin": "1HD1KAE12FB099999", "plateNumber": "TRK-9001", "label": "Heavy Hauler 9001", "model": "Kenworth T680", "year": 2024, "assignedDeviceId": "872312858278", "assignedDriverId": "drv-f4685a30", "assignedDriverName": "Marcus Vance", "odometer": { "value": 145230.8, "unit": "km", "source": "ECU" }, "active": true } ], "page": 0, "size": 50, "totalItems": 1, "totalPages": 1, "last": true }
```


GET /api/v1/integrations/truckercloud/vehicles/{vin} — Vehicle Lookup by 17-Char VIN

Function & Purpose: *Fetches individual vehicle details using its unique 17-character VIN identifier according to ISO 3779 standard.*
Allows TruckerCloud systems to immediately resolve a physical DashCam and current driver assignment from a known chassis VIN.

Parameter / Header	Type	Description
X-API-KEY	*Header*	Required. Integration secret key
vin	*Path*	Required. Validated 17-character vehicle identification number (e.g., 1HD1KAE12FB099999)

Real Response Payload (Sanitized Sample):


```
{ "id": "veh-aebf556b", "vin": "1HD1KAE12FB099999", "plateNumber": "TRK-9001", "label": "Heavy Hauler 9001", "model": "Kenworth T680", "year": 2024, "assignedDeviceId": "872312858278", "assignedDriverId": "drv-f4685a30", "assignedDriverName": "Marcus Vance", "odometer": { "value": 145230.8, "unit": "km", "source": "ECU" }, "active": true }
```

GET /api/v1/integrations/truckercloud/drivers — Fleet Drivers Directory

Function & Purpose: *Provides a paginated list of all active fleet drivers with CDL license information and current vehicle pairings.*
Used by TruckerCloud to attribute driver safety scores, incident investigations, and ELD compliance records to specific drivers.

Parameter / Header	Type	Description
X-API-KEY	*Header*	Required. Integration secret key
page	*Query*	Optional. Page number (default: 0)
size	*Query*	Optional. Page size (default: 50)

Real Response Payload (Sanitized Sample):


```
{ "items": [ { "id": "drv-f4685a30", "name": "Marcus Vance", "phone": "+1-404-555-0188", "email":
"marcus.vance@gbtfleet.com", "licenseNumber": "GA-CDL-882910", "licenseState": "GA", "assignedVehicleId":
"veh-aebf556b", "assignedVehiclePlate": "TRK-9001", "active": true } ], "page": 0, "size": 50, "totalItems": 1,
"totalPages": 1, "last": true }
```


GET /api/v1/integrations/truckercloud/telemetry/latest — Latest Live GPS & Telematics Fix

Function & Purpose: *Returns the most recent live location, speed, bearing, ignition status, and ECU odometer for all active fleet vehicles.*
Supplies TruckerCloud live dispatch maps with real-time location telemetry received over physical JT/T 808 0x0200 packets.

Parameter / Header	Type	Description
X-API-KEY	*Header*	Required. Integration secret key
terminalId	*Query*	Optional. Filter by specific DashCam IMEI / terminal ID

Real Response Payload (Sanitized Sample):


```
[ { "vehicleId": "veh-aebf556b", "vin": "1HD1KAE12FB099999", "licensePlate": "TRK-9001", "imei":
"867543051234567", "terminalId": "872312858278", "timestamp": "2026-11-04T06:15:30Z", "timezone": "UTC",
"driverId": "drv-f4685a30", "driverName": "Marcus Vance", "latitude": 22.746880, "longitude": 75.898868,
"speedKmH": 0.0, "speed": 0.0, "heading": 185.0, "altitudeMeters": 540.0, "ignition": true, "odometer": {
"value": 145230.8, "unit": "km", "source": "ECU" } } ]
```


GET /api/v1/integrations/truckercloud/telemetry/history — Historical GPS Breadcrumbs & Trajectory

Function & Purpose: *Retrieves historical GPS position records supporting vehicle/terminal filtering, start/end UTC dates, and pagination.*
Allows TruckerCloud to perform exact historical route playback, geofence ingress/egress audits, and stop verification.

Parameter / Header	Type	Description
X-API-KEY	*Header*	Required. Integration secret key
vehicleId	*Query*	Optional. Filter by vehicle unique ID (e.g., veh-aebf556b)
terminalId	*Query*	Optional. Filter by DashCam IMEI
startTime	*Query*	Optional. UTC ISO 8601 start time (e.g., 2026-11-04T00:00:00Z)
endTime	*Query*	Optional. UTC ISO 8601 end time (e.g., 2026-11-04T12:00:00Z)
page / size	*Query*	Optional. Pagination parameters (size max: 1000)

Real Response Payload (Sanitized Sample):


```
{ "items": [ { "vehicleId": "veh-aebf556b", "vin": "1HD1KAE12FB099999", "licensePlate": "TRK-9001", "imei":
"867543051234567", "terminalId": "872312858278", "timestamp": "2026-11-04T06:15:30Z", "timezone": "UTC",
"driverId": "drv-f4685a30", "driverName": "Marcus Vance", "latitude": 22.746880, "longitude": 75.898868,
"speedKmH": 0.0, "speed": 0.0, "heading": 185.0, "altitudeMeters": 540.0, "ignition": true, "odometer": {
"value": 145230.8, "unit": "km", "source": "ECU" } } ], "page": 0, "size": 100, "totalItems": 1, "totalPages":
1, "last": true }
```

GET /api/v1/integrations/truckercloud/safety/events — ADAS, DMS & G-Sensor Safety Events

Function & Purpose: *Returns camera-generated safety incidents including harsh braking, rapid acceleration, collision, distraction, and fatigue.*
Includes vehicle and driver association, exact GPS location at time of event, severity score, and video proof availability.

Parameter / Header	Type	Description
X-API-KEY	*Header*	Required. Integration secret key
type	*Query*	Optional. Event type (e.g., HARSH_BRAKE, COLLISION, DISTRACTION)
severity	*Query*	Optional. Severity filter (LOW, MEDIUM, HIGH, CRITICAL)
vehicleId	*Query*	Optional. Filter by vehicle ID
page / size	*Query*	Optional. Pagination controls

Real Response Payload (Sanitized Sample):


```
{ "items": [ { "eventId": "evt-7729104", "vehicleId": "veh-aebf556b", "vin": "1HD1KAE12FB099999",
"licensePlate": "TRK-9001", "imei": "867543051234567", "driverId": "drv-f4685a30", "driverName": "Marcus Vance",
"eventType": "HARSH_BRAKE", "severity": "HIGH", "timestamp": "2026-11-04T04:15:22Z", "timezone": "UTC",
"channel": 1, "latitude": 22.746880, "longitude": 75.898868, "speedKmH": 45.2, "speed": 45.2, "odometer": {
"value": 145230.8, "unit": "km", "source": "ECU" }, "mediaStatus": "READY", "roadVideoUrl":
"https://api.gbttelematics.com/media/incidents/evt-7729104_ch1.m3u8", "cabinVideoUrl":
"https://api.gbttelematics.com/media/incidents/evt-7729104_ch2.m3u8", "mediaExpiresAt": "2026-11-05T04:15:22Z" }
], "page": 0, "size": 50, "totalItems": 1, "totalPages": 1, "last": true }
```


GET /api/v1/safety/recording-url — Incident Video Proof Playback URLs

Function & Purpose: *Generates secure, documented HLS streaming links for road and cabin camera channels linked to a specific safety event ID.*
Allows TruckerCloud portal users to review high-definition dual-camera video evidence with signed link expiration safeguards.

Parameter / Header	Type	Description
X-API-KEY	*Header*	Required. Integration secret key (or Bearer JWT)
eventId	*Query*	Required. Safety event identifier (e.g., evt-7729104)

Real Response Payload (Sanitized Sample):


```
{ "eventId": "evt-7729104", "terminalId": "872312858278", "vehicleId": "veh-aebf556b", "vin":
"1HD1KAE12FB099999", "roadCameraUrl": "https://3.85.81.10/media/incidents/evt-7729104_ch1.m3u8",
"cabinCameraUrl": "https://3.85.81.10/media/incidents/evt-7729104_ch2.m3u8", "mediaStatus": "READY",
"expiresAt": "2026-11-04T16:30:00Z" }
```

GET /api/v1/integrations/truckercloud/trips — Fleet Trips & Mileage Records

Function & Purpose: *Retrieves summarized vehicle trips with departure/arrival timestamps, start/end coordinates, duration, and distance traveled.*
Supplies operational data required by TruckerCloud for fuel consumption estimation, route compliance, and IFTA tax audits.

Parameter / Header	Type	Description
X-API-KEY	*Header*	Required. Integration secret key
vehicleId	*Query*	Optional. Filter trips by vehicle ID
startTime / endTime	*Query*	Optional. Date range filter in UTC ISO 8601
page / size	*Query*	Optional. Pagination parameters

Real Response Payload (Sanitized Sample):


```
{ "items": [ { "id": "trip-5501", "vehicleId": "veh-aebf556b", "vin": "1HD1KAE12FB099999", "driverId":
"drv-f4685a30", "startTime": "2026-11-04T01:00:00Z", "endTime": "2026-11-04T05:00:00Z", "startLocation": {
"lat": 22.7468, "lng": 75.8988, "address": "Indore Fleet Hub" }, "endLocation": { "lat": 23.1765, "lng":
75.7885, "address": "Ujjain Logistics Depot" }, "distanceKm": 58.4, "durationMinutes": 65, "odometerStartKm":
145172.4, "odometerEndKm": 145230.8, "odometerSource": "ECU" } ], "page": 0, "size": 50, "totalItems": 1,
"totalPages": 1, "last": true }
```


POST /api/v1/video/sd/query — Remote MicroSD Card Video Query (JT/T 808 0x9205)

Function & Purpose: *Dispatches an over-the-air JT/T 808 0x9205 instruction to the DashCam to inspect recorded video clips stored on its MicroSD card.*
Enables TruckerCloud to discover all available continuous recording blocks without transferring heavy video data to cloud storage.

Parameter / Header	Type	Description
X-API-KEY	*Header*	Required. Integration secret key
deviceId	*Body JSON*	Required. Terminal IMEI / device ID
channel	*Body JSON*	Required. Camera channel: 1 (Road) or 2 (Cabin)
startTime	*Body JSON*	Required. Query start timestamp in UTC ISO 8601
endTime	*Body JSON*	Required. Query end timestamp in UTC ISO 8601

Real Response Payload (Sanitized Sample):


```
{ "success": true, "deviceId": "872312858278", "channel": 1, "recordings": [ { "channel": 1, "startTime":
"2026-11-04T03:00:00Z", "endTime": "2026-11-04T03:30:00Z", "alarmType": 0, "mediaType": 0, "streamType": 0,
"storageType": 0, "fileSizeBytes": 524288000 } ], "total": 1 }
```

POST /api/v1/video/sd/retrieve — On-Demand Video Clip Retrieval Job (JT/T 808 0x9201)

Function & Purpose: *Creates an asynchronous video retrieval job commanding the DashCam via 0x9201 to upload a specific time clip to cloud storage.*
Guarantees that cloud bandwidth and AWS costs are consumed only for explicitly requested incidents and investigation clips.

Parameter / Header	Type	Description
X-API-KEY	*Header*	Required. Integration secret key
deviceId	*Body JSON*	Required. Terminal IMEI
channel	*Body JSON*	Required. 1 for Road camera, 2 for Cabin camera
startTime	*Body JSON*	Required. UTC ISO 8601 clip start
endTime	*Body JSON*	Required. UTC ISO 8601 clip end

Real Response Payload (Sanitized Sample):


```
{ "jobId": "job-ret-991204", "status": "QUEUED", "deviceId": "872312858278", "channel": 1, "startTime": "2026-11-04T03:10:00Z", "endTime": "2026-11-04T03:15:00Z", "message": "Video retrieval job successfully queued with physical DashCam." }
```


GET /api/v1/video/jobs/{jobId} — Video Retrieval Job Progress & Download URLs

Function & Purpose: *Polls the real-time upload progress and completion status of an on-demand DashCam video extraction job.*
Returns ready HLS playback URLs and MP4 download links once the physical camera completes JT/T 1078 media upload.

Parameter / Header	Type	Description
X-API-KEY	*Header*	Required. Integration secret key
jobId	*Path*	Required. Retrieval job identifier returned by /api/v1/video/sd/retrieve

Real Response Payload (Sanitized Sample):


```
{ "jobId": "job-ret-991204", "status": "COMPLETED", "progressPercent": 100, "deviceId": "872312858278", "channel": 1, "playbackUrl": "https://3.85.81.10/media/retrievals/job-ret-991204.m3u8", "downloadUrl": "https://3.85.81.10/media/retrievals/job-ret-991204.mp4", "completedAt": "2026-11-04T05:35:10Z" }
```


3. TruckerCloud Technical Review Responses & Non-ELD Fleet Architecture

Below are the certified architectural responses to the 9 review items submitted by TruckerCloud Support (Victor Lagos), along with the operational specification for deploying DashCams in standalone environments without an ELD:

Item # & Topic	TruckerCloud Inquiry	GBT Architectural Resolution & Verification
1. Transport Security	Is there/will there be a TLS-secured HTTPS endpoint for production traffic?	**Resolved:** Nginx 1.28 TLS reverse proxy enabled on port 443 with RSA 2048-bit SAN certificate (https://3.85.81.10). Port 80 automatically 301 redirects to HTTPS.
2. Test Environment	Is there a sandbox environment separate from production?	**Resolved:** Dedicated Sandbox key gbt-tc-sandbox-key-7a3e5c12 provided. Validated via /api/v1/integrations/truckercloud/auth/validate which returns environment context.
3. Rate Limits	What are request limits for 5-15 min polling across all vehicles?	**Resolved:** Fully supported. Server capacity configured for up to 1,200 requests/minute per tenant. Polling every 5-15 minutes will experience 0 throttling or rate-limiting.
4. Timezone Field	Can a timezone field be added to /telemetry/latest and /telemetry/history?	**Resolved:** Explicit timezone: "UTC" field added to all telemetry payloads, historical pings, and safety event objects.

5. Safety Odometer	Can /safety/events include vehicle odometer at event time?	**Resolved:** odometer: {"value": 145230.8, "unit": "km", "source": "ECU"} included in every safety event record.
6. Driver Attribution	Can telemetry payloads include driverId/driverName directly?	**Resolved:** Direct driverId and driverName columns added to telemetry entity and responses, reflecting driver assigned at the time of each ping.
7. Data Retention	What is the maximum retention period? Minimum 90 days needed.	**Resolved:** Database retention guaranteed for a minimum of 180 days (surpassing 90-day requirement). Partitioned GPS history tables preserve full breadcrumb trails.
8. Speed Units	Is speed always in km/h consistent with speedKmH in safety events?	**Resolved:** Standardized to km/h across all endpoints. Payloads provide both speedKmH and speed aliases for seamless compatibility.
9. API Key Rotation	Rotate live production key and deliver through secure channel.	**Resolved:** Production key rotated to gbt-tc-prod-key-9f82d14b. Future production keys delivered out-of-band via secure encrypted channel.

Standalone DashCam (Non-ELD) Administration & UI Controls

When DashCams are installed in commercial vehicles without an electronic logging device (ELD) or J1939 ECU connector, the platform provides dedicated fleet endpoints and UI controls in the mobile/web app to ensure full operational compliance:
• **Add Vehicle (POST /api/v1/vehicles):** Fleet managers register vehicle profile with VIN, license plate, make, and initial odometer.
• **Add Driver (POST /api/v1/drivers):** Drivers are registered with legal name, phone, and commercial driver license number.
• **Assign / Change Driver (POST /api/v1/vehicles/{id}/assign-driver):** Admins can reassign drivers on-the-fly either by registered driverId or ad-hoc driverName.
• **Odometer Provenance:** In standalone mode, odometer readings are labeled source: "GPS_ESTIMATED", calculated from GPS geodesic distance accumulation from initial baseline, ensuring full data integrity without fabricating ECU bus readings.

4. Integration Acceptance & Support Verification

Live Production Validation: All endpoints in this specification have been validated against our live AWS EC2 production deployment (https://3.85.81.10) secured with TLS 1.2/1.3 and backed by MySQL 8.4 persistence and active WanWay H20P JT/T 808/1078 gateway services. All TruckerCloud acceptance tests completed with a 100% pass rate.

Interactive Testing:
• **Live Swagger UI:** https://3.85.81.10/swagger-ui/index.html
• **Raw OpenAPI 3.0 Spec:** https://3.85.81.10/v3/api-docs
• **Postman Collection:** A pre-configured collection file GBT_TruckerCloud_Postman_Collection.json is provided in the repository with real, sanitized requests and pre-set X-API-KEY variables.

Commercial Terms: One-time integration fee of $15,000 accepted; completion timeline target end of February 2027 confirmed.
Technical Integration Contact: GBT Technologies Engineering Team • Email: api-support@gbt.com